Australian Data Archive (ADA) and Australian National University (ANU) and The Dataverse Project, TKLabels use case

Janet McDougall, Steven McEachern - Australian Data Archive, ANU
Sonia Barbosa -The Dataverse Project, Harvard University

Open Repositories Conference 2023, Cape Town SA

Generalist repositories and indigenous knowledge

- Many archives have large holdings of content about (and from) Indigenous communities and people
- Increasing recognition of the need to understand, include and represent indigenous forms of knowledge and rights in repositories and archives
- This presentation is an introduction to work currently occurring in the Dataverse community to support this
- Key aim is to recognise and enable indigenous data sovereignty: **“The right of Indigenous Peoples and nations to govern the collection, ownership, and application of their own data.”**

WHAT ARE **INDIGENOUS DATA?**

Data, information and knowledges, in any format, that impacts Indigenous Peoples, nations, and communities at the collective and individual levels:

DATA ABOUT OUR NON-HUMAN RELATIONS

Land, water, geology, titles, air, soil, sacred ecosystems, territories, plants, animals, etc.

DATA ABOUT US AS INDIVIDUALS

Administrative, legal, health, social, commercial, corporate, services, etc.

DATA ABOUT US AS COLLECTIVES

Traditional and cultural information, languages knowledge systems, ancestral and clan knowledges, etc.

USINDIGENOUSDATA.ORG
@USIDSN

Informed by British Columbia First Nations Data Governance Institute - BCFNDGI.COM

GIDA-GLOBAL.ORG
@GidaGlobal

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

INDIGENOUS DATA SOVEREIGNTY

The *right* of Indigenous Peoples and nations to govern the collection, ownership, and application of their own data.

1. Derives from inherent rights to govern peoples, lands, and resources.
2. Genesis in traditions, roles, and responsibilities for the use of community held information.
3. Positioned within a human rights framework and court cases, treaties, and/or recognition.
4. Knowledge belongs to the collective and is fundamental to who we are as peoples.

See Kukutai T & Taylor J. (Eds). (2016). Indigenous Data Sovereignty. Canberra: Australian National University Press.

USINDIGENOUSDATA.ORG | @USIDSN

GIDA-Global.ORG | @GIDAGlobal

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

Collective Benefit Data ecosystems shall be designed and function in ways that enable Indigenous Peoples to derive benefit from the data.

Authority to Control Indigenous Peoples' rights and interests in Indigenous data must be recognised and their authority to control such data be empowered. Indigenous data governance enables Indigenous Peoples and governing bodies to determine how Indigenous Peoples, as well as Indigenous lands, territories, resources, knowledges and geographical indicators, are represented and identified within data.

Responsibility Those working with Indigenous data have a responsibility to share how those data are used to support Indigenous Peoples' self determination and collective benefit. Accountability requires meaningful and openly available evidence of these efforts and the benefits accruing to Indigenous Peoples.

Ethics Indigenous Peoples' rights and wellbeing should be the primary concern at all stages of the data life cycle and across the data ecosystem.

Research Data Alliance International Indigenous Data Sovereignty Interest Group. (September 2019). "CARE Principles for Indigenous Data Governance." The Global Indigenous Data Alliance. GIDA-global.org

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

https://www.gida-global.org/care

About Local Contexts

- Local Contexts is a global initiative that supports Indigenous communities with tools that attribute cultural authority of heritage and data. By focusing on Indigenous Cultural and Intellectual Property and Indigenous Data Sovereignty, Local Contexts helps Indigenous communities repatriate knowledge and gain control over how their data is collected, managed, displayed, accessed, and used in the future.

TK/BC Labels

Indigenous communities

- reinforce rights by applying TK and BC Labels
- Traditional Knowledge (TK) and Biocultural (BC) Labels establish Indigenous cultural authority and governance over Indigenous data and collections by adding provenance information and contextual metadata (including community names), protocols, and permissions for access, use, and circulation.
- The Label descriptions can be customized by Indigenous communities to ensure that their local and traditional protocols are embedded in those Labels.
- When Indigenous communities apply those Labels to relevant Projects, the Labels will be used as an identifier to define how their cultural material is accessed and used

Institutions and researchers

- disclose interests by placing Notices
- Notices are tools for institutions, repositories, and researchers to support ethical use and reinforce relationships with Indigenous communities. They also work to educate the public around Indigenous rights and interests in historical and future collections and data.
- Institutions and researchers can create Projects for cultural materials they protect and apply Notices to them.
- By applying Notices it will notify Indigenous communities of Indigenous rights and interests in these collections and data.

Presenting and representing indigenous knowledge

Systematics Collectio

CHR 559987 A – Gentianella lute

Data provider:	Allan Herbarium
Barcode:	CHR 559987 A
Type status:	Holotype
Specimen type:	Sheet
Database record added:	13 June 2002
Database record updated:	24 February 2023

Components

Primary component

Active identification

Habitat:	Blowout in granite sand on ridge, with Chionohebe pulvinaris Dracophyllum pronum, Poa colensoi, Luzula pumila.
Notes:	Monocarpic, perhaps biennial or triennial but some plants flowering each year. Pollinator seen: bumble bee. Common from this point north along the range, abundant wherever there is bare granite sand. Gentianella "Lookout".Specimen over 2 sheets.

Specimen notes

Type:	HOLOTYPE of Gentianella luteoalba Glenny. New Zealand Journal of Botany 42(3): 361-530 (2004).

Permissions

Project permits

Project title:
Local Contexts - Allan Herbarium (CHR)

Reference:
CHR Collection - Local Contexts

BC Biocultural (BC) Notice

Terms of use | Feedback & enquiries

© 2013 - 2023 Version 2.3.0.1323

Manaaki Whenua Landcare

https://scd.landcareresearch.co.nz/Specimen/CHR%20559987%20A

About the Dataverse Project

- The Dataverse Project is an open source web application to share, preserve, cite, explore, and analyze research data. It facilitates making data available to others, and allows you to replicate others' work more easily. Researchers, journals, data authors, publishers, data distributors, and affiliated institutions all receive academic credit and web visibility.
- A Dataverse repository is the software installation, which then hosts multiple virtual archives called Dataverse collections. Each Dataverse collection contains datasets, and each dataset contains descriptive metadata and data files (including documentation and code that accompany the data).
- The central insight behind the Dataverse Project is to automate much of the job of the professional archivist, and to provide services for and to distribute credit to the data creator.
- https://dataverse.org/about

99 Installations

How do we represent indigenous knowledge and rights in the Dataverse platform?

Local Contexts and TC/BC Labels and Notices: Displaying Notices

- The text, icons, and titles for the Notices cannot be changed.
- The only changes that can be made to the descriptive text is replacing the placeholder “our institution” with the name of the institution displaying the Institution Notices.
- **The Notices are intended to be displayed prominently** on public-facing institutional websites, on digital collections pages, and or in finding aids.
- **When displaying the Notices, the title, icon, and text must all be displayed**.
- The Notices should be applied at a metadata field level.

Dataverse:

- DDI based
- Capable of presenting broad range of metadata
- Includes licensing, discovery metadata, access controls, …

- BUT:
- Designed for representation of metadata consistent with traditional notions of rights, knowledge and access
- How do we enable and appropriately reflect indigenous knowledge and rights within a “traditional” repository framework

What would we need to incorporate?

- Representation of knowledge (e.g. TK/BC Labels)
- Recognition of indigenous ownership rights (data access)
- Curation of knowledge
- Presentation of this information in the Dataverse portal

The approach

- Established discussion between Dataverse, Local Contexts and active indigenous and non-indigenous service providers, researchers and developers, following the 2022 Dataverse Community Meeting
- Series of meetings from September 2022 to present, looking at scoping of possible project requirements and use cases
  - Included case from the ADA associated with the Indigenous Data Network project in Australia
- Opportunity to pilot some integration as part of the NIH GREI program, of which Dataverse is a part
- Proposal currently in development with NIH through Harvard as Dataverse project leads

Pilot tool (pre-project)

- RIDAGOP toolkit: https://toolkit.ridagop.net
- Developed by Ramin Soleymani-Fard and the team at the Social-Ecological Systems Research Group, Laboratory for the Analysis of Social-Ecological Systems in a Globalised world (LASEG) is a research group at the Institute of Environmental Science and Technology, Universitat Autònoma de Barcelona (ICTA-UAB)
- Collaboration with Local Contexts
- Demonstration of toolkit: https://dataverse.org/dataversetv, April 2023

Assign Local Contexts Labels to dataverse datasets

Local Contexts

The Dataverse Project

Local Contexts Project ID

Copy the Project ID of your Local Contexts Project, from which you want to use the labels. The ID can be taken from the Identifiers section of the project description or the project address

https://localcontextshub.org/projects/ e1ab3fa-120a-127c-94e2-b7d23ce61878

Fetch Labels

▼ See the project details here, once you fetched them

Dataverse details

Please specify the dataverse instance (e.g. https://data.harvard.edu/dataverse), the repository DOI (which you find in the metadata) and the API key

Dataverse instance: https://data.harvard.edu/dataverse

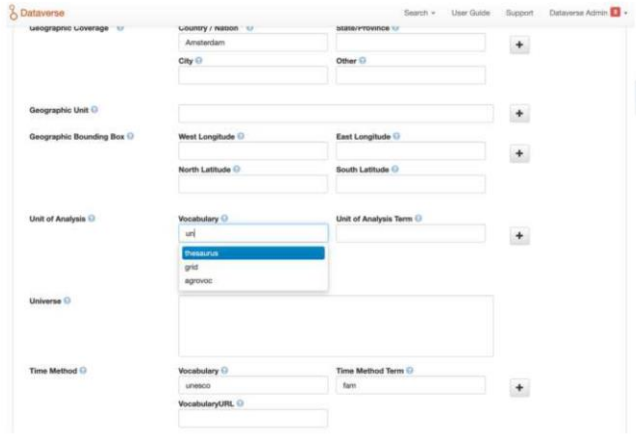
DOI: https://data.harvard.edu/dataverse

API Key:

Read dataset

Integrated with Dataverse External CV support

Dataverse deposit form with selected CVs

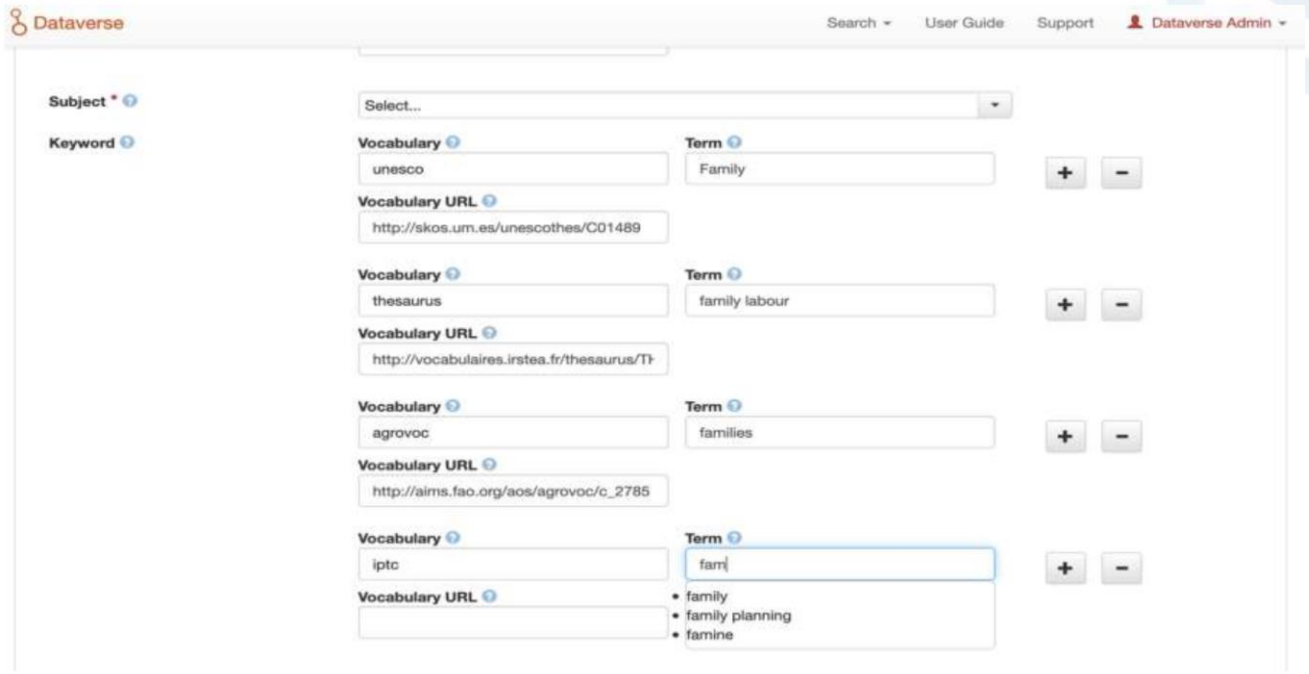


Every field could be linked to the controlled vocabularies in FAIR way!

This project is funded from the EU Horizon 2020 Research and Innovation Programme (2014-2020) under Grant Agreement No. 823782

SSHOC

One metadata field linked to many ontologies



Language switch in Dataverse will change the language of suggested terms!

Source: Tykhonov, 2021, DV Community Meeting

Next steps

- Ongoing engagement with NIH on funding for pilot project
- Scoping of requirements
  - Integration methods for connecting Dataverse to Local Contexts (via LC API services) and
  - Suitable user interface for presentation
  - Curation workflows for including labels and notices
    - That align with community and repository practices and capabilities
  - How to align to notices from specific communities
- Proposal to NIH expected in second half of 2023
  - More on this from Sonia